

Project ID :

25-26J-179

1. Topic (12 words max)

Machine Learning Applications in Financial Leasing

2. Research group the project belongs to

SST - Software Systems & Technologies

3. Specialization of the project belongs to

Information Technology (IT)

4. If a continuation of a previous project:

Project ID	
Year	

5. Brief description of the research problem including references (200 – 500 words max) – references not included in word count.

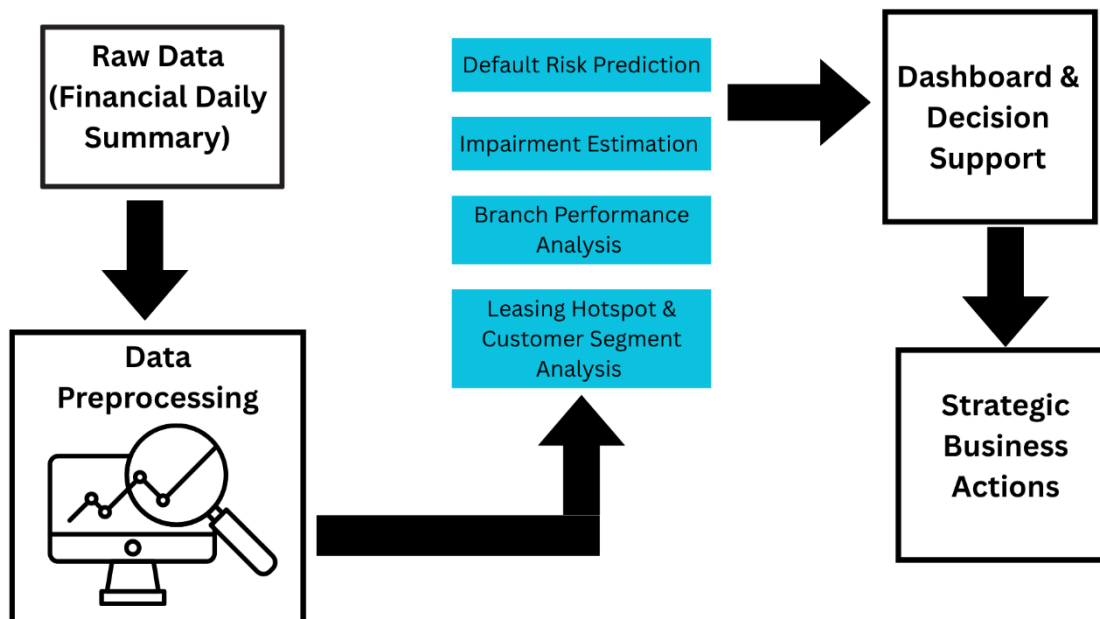
The management of customer default risk is a significant issue in the financial leasing sector, with direct implications for profitability, capital resilience, and overall financial soundness. Existing credit evaluation models, primarily based on past financial data, often fail to capture behavioral trends that may indicate rising default risk [2], [6]. As a result, these models may delay the identification of high-risk clients, contributing to an increase in non-performing loans (NPLs). In a fast-paced and uncertain market environment, leasing companies need advanced tools that provide quick, precise, and adaptive credit assessments. This research proposes a modern decision support system built on machine learning methods. It employs classification, regression, and clustering algorithms to address four key areas: default risk prediction, expected credit loss estimation, branch performance evaluation, and the identification of leasing hotspots and high-value customer segments [4]. By integrating financial metrics with behavioral patterns, the system significantly optimizes resource allocation and enables the early identification of potential default risks [1], [6]. Furthermore, the advanced accuracy, clarity, and explainability provided by modern machine learning techniques especially those incorporating explainable AI substantially enhance the system's ability to deliver timely, data driven realization[4]. This in turn, allows financial institutions to adopt more proactive and effective risk management practices, while also fostering greater operational efficiency and ensuring long-term business sustainability [3], [5].

References

- [1]M. G. Abebe, "The effect of asset and liability management on the financial performance of microfinance institutions: evidence from sub-Saharan African region," *Future Business Journal*, vol. 8, no. 1, 2022. [Online]. Available: <https://doi.org/10.1186/s43093-022-00134-8>.
- [2]R. Nyman, S. Kapadia, and D. Tuckett, "News and narratives in financial systems: Exploiting big data for systemic risk assessment," *Journal of Economic Dynamics & Control*, vol. 127, 2021. [Online]. Available: <https://doi.org/10.1016/j.jedc.2021.104119>.
- [3]S. Sihombing and M. Khaddafi, "Luxury car leasing," *Article*, 2024. [Online]. Available: <https://doi.org/10.46799/adv.v2i6.264>.
- [4]P. Appiahene, Y. M. Missah, and U. Najim, "Predicting bank operational efficiency using machine learning algorithm: Comparative study of decision tree, random forest, and neural networks," *Advances in Fuzzy Systems*, vol. 2020, Article ID 8581202, 2020. [Online]. Available: <https://doi.org/10.1155/2020/8581202>.
- [5] L. Pierce, "Organizational structure and the limits of knowledge sharing: Incentive conflict and agency in car leasing," *Management Science*, vol. 68, no. 4, pp. 1822–1842, 2022. [Online]. Available: <https://doi.org/10.1287/mnsc.1110.1472>.

6. Brief description of the nature of the solution including a conceptual diagram (250 words max)

Creating a machine learning-based decision support system for the financial leasing sector is the goal of this research project. We found four essential elements that tackle important business issues after carefully examining the dataset that was made public. Among these are: (1) Default Risk Prediction, which uses classification models to identify customers who are likely to stop making payments; (2) Impairment Estimation, which forecasts expected credit loss (ECL) on leasing facilities using regression techniques; (3) Branch Performance Analysis, which considers operational efficiency across locations using clustering and aggregation; and (4) Leasing Hotspot and Customer Segment Analysis, which uses clustering and classification to identify profitable regions and customer profiles. By combining these models into a single, data-driven framework, proactive credit risk management, better financial provisioning, and more astute strategic planning will be made possible, enabling the company to grow sustainably in a cutthroat financial environment.



7. Brief description of specialized domain expertise, knowledge, and data requirements (300 words max)

This project requires specific expertise in Time Series Analysis (ARIMA) and financial leasing. Accurately defining issue statements and interpreting model outputs require a thorough grasp of financial leasing activities, including credit risk assessment, non-performing loans (NPL), and impairment estimation. Along with proficiency in data preprocessing, feature engineering, and model evaluation, proficiency in machine learning techniques like classification, regression, and clustering is equally crucial. Customer profiles, lease account information, transaction histories, repayment patterns, and branch-level performance statistics are among the comprehensive datasets that the financial institution provides for the project. To create accurate and trustworthy models, historical data must be clean, thorough, and sizable enough. The system will provide actionable insights to help strategic decision-making in the financial leasing sector if domain expertise and data-driven modeling are successfully integrated.

8. Objectives and Novelty

Main Objective To develop an intelligent, machine learning-based decision assistance system that enhances strategic decision-making in the financial leasing industry by accurately predicting risks, improving financial provisioning, evaluating operational performance, and uncovering market opportunities.			
Member Name with Registration No	Sub Objective	Tasks	Novelty
Semasingha W. A. L. D. IT22198518	To develop a regression-based machine learning model that accurately estimates the expected credit loss (impairment) on leasing facilities, in line with financial risk management	- Understand impairment calculation logic based on financial domain knowledge. - Preprocess the dataset by cleaning, handling missing values, and generating relevant financial features. - Select and engineer features that influence impairment, such as delinquency history, customer risk profile, lease type, and economic conditions. - Train and evaluate regression models using appropriate performance metrics like RMSE or MAE.	Uses machine learning for impairment estimation, which traditionally relies on static rule-based financial models.

		• Interpret model predictions using SHAP values or feature importance to ensure transparency and compliance. • Validate predictions against historical impairment values or expert-estimated provisions. • Integrate results into a dashboard for visualization and business decision support.	
Maharambage K.N IT22129130	Default Risk Prediction (Predicting default risk in vehicles leasing) -: Creating a precise and interpretable machine learning model to forecast default risk in vehicle leasing clients by utilizing financial and behavioral information to facilitate forward-looking	• Gather customers' financial records as well as their behavioral data from the leasing company's databases. • Prepare, clean, and preprocess raw data for its use in machine learning models. • Create new features like lagging payments, employment history stability, and debt-to-income ratios. • Train various classification models. Integrate the selected model into a web-based decision support system for real-time default prediction.	• Behavioral-Enriched Features: Melding behavioral elements like overdue payments, partial payment submissions and irregular transactions with credit risk scoring. • Dynamic Feature Engineering: Incorporating temporal aspect of trends in history of a particular lease to improve prediction accuracy for defaults as time progresses. • Customized Data Cleaning Pipelines: Specialized workflows tailored toward

	management of credit risk.		specific domains for recurrent gaps, inconsistency or delayed capture of client's financial data automations
Weerasinghe M.P IT22087560	Branch Performance Analysis	• Collect operational data at the branch level, including service time, payback rates, lease volume, client acquisition, and employee efficiency. • To ensure correctness, consistency, completeness, cleanliness and process unstructured branch data. • To categorize branches according to integrated performance measures and use clustering techniques like KNN. • Use statistical aggregation and scoring to rank branches based on the key metrics like revenue growth, the non-performing loan ratio, and customer engagement.	Combining real-time performance visualization with supervised learning techniques makes branch operation analysis more intelligent and dynamic. The system can simultaneously monitor and display critical operational metrics and use machine learning algorithms like clustering and anomaly detection to uncover hidden trends, inefficiencies, and performance segments that may not be apparent through traditional analysis techniques. In order to provide decision makers with timely, data-driven views that enable improved strategic planning and targeted performance improvement across branches. This strategy merges algorithmically generated insights with visual analytics. It promotes ongoing operational excellence, increases transparency and it increases resource efficiency also.

		Develop a branch performance monitoring dashboard to be included within a web application to track performance in real time.	
Erangi M G M IT22583796	Leasing Hotspot & Customer Segment Analysis- This is useful for identifying the best locations to open leasing branches and target customer profiles to expand leasing operations.	• Collect and preprocess past data on leasing transactions, customer profile and branch performance. • Clean and transform data to prepare it for analysis and modeling. • Apply clustering techniques to identify leasing hotspots and group geographic areas based on leasing potential. • Apply classification models to segment customers and predict the chance of future leasing behavior. • Produce informed predictions and recommendations to drive effective strategies for branch development and customer engagement.	This component introduces the creativity use of clustering and classification techniques combined with geospatial analysis to strategically guide branch expansion in the Sri Lankan financial leasing sector. The project enhances traditional business analysis by integrating predictive models with geographic insights to produce precise, data-driven recommendations. It provides leasing companies with a powerful framework to enhance branch placement and customer targeting, leveraging advanced analytics to drive future leasing success.

9. Individual component description of how it is complied with the specialization.

Member Name with Registration No	Description
Semasingha W. A. L. D. IT22198518	Predictive modeling, data analytics, and machine learning are examples of key IT expertise areas that are applied in this component. In order to quantify financial damage, regression models are created using programs like Python and scikit-learn. Key IT competencies are reflected in tasks like feature engineering, data preprocessing, and model validation. The results' incorporation into a decision-support system is also consistent with the degree's emphasis on software development and system integration.
Maharambage K.N IT22129130	This advanced technology aims to identify critical gaps in early credit risk detection, therefore dealing with financial risk management. The goal is to build a predictive model that will enable vehicle lease companies to gauge a customer's default probability before approving a lease contract so that the company can avoid incurring losses from non-performing loans. Thus, the analytics system's precision combines behavioral data with financial data and goes beyond basic scoring models. Solutions, this guarantees real-time explainable risks giving pre-computed scenarios which aid compliance-driven decisions while upholding regulatory guidelines. The system strengthens preset limits on predictive estimates and allows real-time portfolio supervision which improves reactive management during high-risk scenario prediction thus optimizing the value sustainability of leasing companies.
Weerasinghe M.P IT22087560	The Branch Performance Analysis tool uses clustering and aggregation techniques to identify patterns and classify financial leasing branches based on performance data, as a result, tracking the operational effectiveness of these branches. This analysis supports strategic decision-making, resource allocation, and performance improvement initiatives. Demonstrating how IT based solutions, such as data analytics, algorithmic modeling, and performance monitoring systems, are applied to actual business problems, it supports the information technology specialization. In order to increase operational effectiveness, it also emphasizes critical skills in data processing, system design, and technology integration.
Erangi M G M IT22583796	This component supports decision-making in the financial leasing sector by applying expertise in database management, information systems, and machine learning. It uses clustering and classification techniques to analyze historical data and predict optimal locations and target customers for branch expansion in Sri Lanka. The research display how predictive analytics can enhance operational efficiency and guide business growth through data driven

	understanding.
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10. Supervisor details

	Title	First Name	Last Name	Signature
Supervisor	Mrs.	Suriya	Kumari	
Co-Supervisor	Ms.	Dulani	Jayasinghe	
External Supervisor	Mr.	Kapila	Premachandra	
Summary of external supervisor's (if any) experience and expertise				

This part is to be filled by the Topic Screening Staff members.

- a) Does the chosen research topic possess a comprehensive scope suitable for a final-year project?

Yes		No	
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- b) Does the proposed topic exhibit novelty?

Yes		No	
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- c) Do you believe they have the capability to successfully execute the proposed project?

Yes		No	
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- d) Do the proposed sub-objectives reflect the students' areas of specialization?

Yes		No	
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- e) Supervisor's Evaluation and Recommendation for the Research topic:

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Acceptable: Mark/Select as necessary

Topic Assessment Accepted	
Topic Assessment Accepted with minor changes*	
Topic Assessment to be Resubmitted with major changes*	
Topic Assessment Rejected. Topic must be changed	

* Detailed comments given below

Comments

Staff Member's Name	Signature

***Important:**

1. According to the comments given by the evaluator, make the necessary modifications and get the approval by the **Evaluator**.
2. If the project topic is rejected, identify a new topic, and request the RP Team for a new topic assessment.